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United States Patent Application Publication

20250284178

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SATO; KATSUHIKO

LIGHT-AMOUNT ADJUSTING APPARATUS, LENS APPARATUS, AND IMAGE PICKUP APPARATUS

Abstract

A light-amount adjusting apparatus includes a drive unit configured to drive blade members, a first opening/closing unit including three or more blade members to open and close at least a part of a first opening according to driving of the drive unit, and a second opening/closing unit including blade members as many as the blade members constituting the first opening/closing unit to open and close at least a part of a second opening according to driving of the drive unit. The first and second opening/closing unit are arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.

Inventors: SATO; KATSUHIKO (Tochigi, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 1000008494737

Appl. No.: 19/066570

Filed: February 28, 2025

Foreign Application Priority Data

JP 2024-035996

Mar. 08, 2024

Publication Classification

Int. Cl.: G03B9/06 (20210101); G03B35/08 (20210101); H04N13/239 (20180101); H04N23/55 (20230101); H04N23/75 (20230101)

U.S. Cl.:

Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a light-amount adjusting apparatus having a compound-eyes optical system.

Description of Related Art

[0002] Stereoscopic image pickup apparatuses and VR video capture apparatuses that perform imaging using a compound-eyes optical system including a plurality of optical systems have conventionally been known. Japanese Patent Laid-Open No. 2014-119615 discloses a light-amount adjusting apparatus configured to adjust a light amount of each optical system by opening and closing a plurality of blade members.

[0003] In the light-amount adjusting apparatus disclosed in Japanese Patent Laid-Open No. 2014-119615, the number of blade members is one or two for each opening, so the obtained blur shape is a polygonal shape with a small number of sides.

SUMMARY

[0004] A light-amount adjusting apparatus according to one aspect of the disclosure is configured to adjust a light amount of a first optical system and a light amount of a second optical system. The light-amount adjusting apparatus includes a drive unit configured to drive blade members, a first opening/closing unit including three or more blade members and configured to open and close at least a part of a first opening through which light is to pass, according to driving of the drive unit, and a second opening/closing unit including blade members as many as the blade members constituting the first opening/closing unit and configured to open and close at least a part of a second opening through which light is to pass, according to driving of the drive unit. The first opening/closing unit and the second opening/closing unit are arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees. A lens apparatus and an image pickup apparatus each having the above light-amount adjusting apparatus also constitute another aspect of the disclosure.

[0005] Further features of various embodiments of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram of an image pickup apparatus according to a first embodiment.

[0007] FIG. 2 is a perspective view of a light-amount adjusting apparatus according to the first embodiment.

[0008] FIG. 3 is an exploded perspective view of the light-amount adjusting apparatus according to the first embodiment.

[0009] FIGS. 4A and 4B explain a rotating direction of a blade drive member and the phase of the blade members in the light-amount adjusting apparatus according to the first embodiment.

[0010] FIGS. 5A and 5B explain the generation principle of a light striation.

[0011] FIG. 6 illustrates light striations.

[0012] FIGS. 7A and 7B explain a difference in the state of light striations according to a shape of an aperture.

[0013] FIGS. 8A and 8B explain light striations generated in a compound-eyes optical system unit according to the first embodiment.

[0014] FIGS. 9A and 9B illustrate example light striations generated in a case where a shape of an aperture in a second embodiment is an odd polygon.

[0015] FIGS. 10A and 10B illustrate example light striations generated in a case where a shape of an aperture in the second embodiment is an even polygon.

[0016] FIGS. 11A and 11B illustrate other examples of light striations generated in a case where a shape of an aperture in the second embodiment is an odd polygon.

[0017] FIGS. 12A and 12B explain the configuration of a light-amount adjusting apparatus according to a third embodiment.

[0018] FIGS. 13A and 13B explain the configuration of a light-amount adjusting apparatus according to a fourth embodiment.

DETAILED DESCRIPTION

[0019] Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.

First Embodiment

[0020] FIG. 1 is a block diagram of an image pickup apparatus **10** according to this embodiment. The image pickup apparatus **10** includes a camera body **1** and a compound-eyes optical system (lens apparatus) **20** including a first optical system **2** and a second optical system **3**. The compound-eyes optical system unit **20** may be fixed to the camera body **1**, or may be attachable to and detachable from the camera body **1** like an interchangeable lens.

[0021] The first optical system **2** and the second optical system **3** include imaging optical systems **201** and **301** including lens units that determine a focal length and focus state and have the same configurations. Each of the imaging optical systems **201** and **301** includes a light-amount adjusting apparatus **100**. The imaging optical systems **201** and **301** may include a single lens or a set of multiple lenses that integrally move in adjusting the focal length and focus position.

[0022] The light-amount adjusting apparatus **100** includes a plurality of blade members (not illustrated) and an opening/closing mechanism (not illustrated) that opens and closes the plurality of blade members, and adjusts a light amount of each optical system. The opening/closing mechanism in the light-amount adjusting apparatus **100** is driven by a light-amount adjusting drive unit (drive unit) **105**. The light-amount adjusting apparatus **100** is a so-called iris diaphragm in which the plurality of blade members arranged around the optical axis partially overlap each other to form an aperture (diaphragm aperture) on the optical axis. In this embodiment, since each of the first optical system **2** and the second optical system **3** has the optical axis, the optical axis of the first optical system **2** will be referred to as OA, and the optical axis of the second optical system **3** will be referred to as OB.

[0023] An aperture value (F-number) increases or decreases according to the positions of the plurality of blade members. An overlap amount of the plurality of blade members also changes according to the positions of the plurality of blade members, and the operating load applied to the light-amount adjusting drive unit **105** changes accordingly. In general, the operating load increases as the aperture value, i.e., the overlap amount of the plurality of blade members, increases.

[0024] The light-amount adjusting drive unit **105** is configured with a stepping motor and is controlled by one of lens control units **204** and **304** provided in the first optical system **2** and the second optical system **3**, respectively. More specifically, one of the lens control units **204** and **304** controls a drive direction of the light-amount adjusting drive unit **105** by changing the polarity of a

drive signal applied to the light-amount adjusting drive unit **105**, and controls a drive position of the light-amount adjusting drive unit **105** by increasing or decreasing the number of pulses of the drive signal. Thereby, an opening and closing operation amount (aperture amount, aperture diameter) of the plurality of blade members in the light-amount adjusting apparatus **100** can be controlled. The light-amount adjusting drive unit **105** may be provided with an aperture position detector (not illustrated) configured to detect the positions of the plurality of blade members corresponding to the maximum aperture.

[0025] The lens control units **204** and **304** control the aperture amount (aperture diameter) of the light-amount adjusting apparatus **100** based on a signal (target aperture-value signal) received from the camera body **1** via lens communication (COMM) units **205** and **305**. The lens control units **204** and **304** also control the aperture diameter of the light-amount adjusting apparatus **100** based on a drive instruction control amount corresponding to the luminance (aperture value).

[0026] The camera body **1** includes a first image sensor **12**, a second image sensor **13**, a camera control unit **14**, and camera communication units **15** and **16**. Each of the first and second image sensors **12** and **13** include a CMOS sensor or a CCD sensor, and performs photoelectric conversion for an optical image (object image) formed via a corresponding one of the first optical system **2** and the second optical system **3** to output image data. In this embodiment, an image sensor is provided for each imaging optical system, but a single image sensor may be configured to output a plurality of image data.

[0027] The camera control unit **14** controls the first image sensor **12**, the second image sensor **13**, and the camera communication units **15** and **16**. The camera control unit **14** transmits information on the target aperture value of the light-amount adjusting apparatus **100** to the lens control units **204** and **304** via the camera communication units **15** and **16** and the lens communication units **205** and **305**.

[0028] FIG. **2** is a perspective view of the light-amount adjusting apparatus **100**. FIG. **3** is an exploded perspective view of the light-amount adjusting apparatus **100**. The light-amount adjusting apparatus **100** includes an aperture base plate (base member) **101**, a first aperture drive member (first drive member) **120**, a second aperture drive ring (second drive member) **121**, a cam plate (press member) **103**, a plurality of blade members **104**, and a light-amount adjusting drive unit **105**.

[0029] The aperture base plate **101** includes a first opening **101a** that has a center on the optical axis OA of the first optical system **2**, and a second opening **101b** that has a center on the optical axis OB of the second optical system **3**. In the first opening **101a**, a plurality of engagement receivers **101c** for holding engagement diameter portions **120a** provided on the first aperture drive ring **120** are evenly arranged along the circumferential direction. The first aperture drive ring **120** is rotatably engaged with (inserted or fitted into) the first opening **101a**. In the second opening **101b**, a plurality of engagement receivers **101d** for holding engagement diameter portions **121a** provided on the second aperture drive ring **121** are evenly arranged along the circumferential direction. The second aperture drive ring **121** is rotatably engaged with (inserted or fitted into) the second opening **101b**. The aperture base plate **101** has a bearing reference hole **101e** provided at a predetermined position with respect to the central axis. A bearing **105b** coaxially and rotatably supports a rotational drive shaft **105a** of the light-amount adjusting drive unit **105**. The bearing **105b** is inserted into the bearing reference hole **101e**, and the fixed portion **105c** provided on the light-amount adjusting drive unit **105** is fixed to the aperture base plate **101** with a screw or the like, so that the rotational drive shaft **105a** is disposed at a predetermined position.

[0030] A pinion gear (transmission member) **106** is fixed to the rotational drive shaft **105a**. A first ring gear (first transmitter) **120b** is provided on the circumference of the first aperture drive ring **120**, which is engaged (or geared) with the pinion gear **106** and rotates the first aperture drive ring **120** by receiving the rotational drive force of the light-amount adjusting drive unit **105**. A second ring gear (second transmitter) **121b** is provided on the circumference of the second aperture drive ring **121**, which is engaged (or geared) with the pinion gear **106** and rotates the second aperture

drive ring **121** by receiving the rotational drive force of the light-amount adjusting drive unit **105**. Each of the first and second aperture drive rings **120** and **121** rotates in association with the rotational drive of the pinion gear **106**, with the central axis coinciding with the optical axis OA and OB as the rotation center. In this embodiment, a single drive unit is provided to drive the first and second aperture drive rings **120** and **121**, but two drive units may be provided to drive the first and second aperture drive rings **120** and **121**, respectively.

[0031] The cam plate **103** is supported by the aperture base plate **101** while an operating space is secured for the plurality of blade members **104** and the first and second aperture drive rings **120** and **121**. The cam plate **103** is provided with a first hole portion **103a** that opens coaxially with the first opening **101a**, and a second hole portion **103b** that opens coaxially with the second opening **101b**.

[0032] The blade members **104** are common parts that are equally provided in number (three or more) along the circumferential direction around the respective rotation axis centers of the first and second aperture drive rings **120** and **121**. The blade members **104** that are arranged along the circumferential direction of the first aperture drive ring **120** function as the first opening/closing unit that opens and closes at least a portion of the first opening **101a** according to the drive of the light-amount adjusting drive unit **105**. The blade members **104** that are arranged along the circumferential direction of the second aperture drive ring **121** function as the second opening/closing unit that opens and closes at least a part of the second opening **101b** according to the drive of the light-amount adjusting drive unit **105**. The first and second aperture drive rings **120** and **121** and the plurality of blade members **104** are housed between the aperture base plate **101** and the cam plate **103**. The blade members **104** are provided with a rotation pin (first engagement member) **104a** that serves as the center of rotation and a cam pin (second engagement member) **104b** that rotates the blade members **104**. The rotation pin **104a** and the cam pin **104b** may have a hole shape.

[0033] The first aperture drive ring **120** has rotating holes **120c** into which the rotation pins **104a** are inserted (or engaged), and the rotating holes **120c** are evenly provided along the circumferential direction. The second aperture drive ring **121** has rotating holes **121c** into which the rotation pins **104a** are inserted, and the rotating holes **121c** are evenly provided along the circumferential direction. Each blade member **104** is supported rotatably about the rotation pins **104a**.

[0034] The cam plate **103** has cam grooves (cam shape) **103c** with which the cam pins **104b** are engaged, which are evenly provided along the circumferential direction and coaxial with the central axis of the first hole portion **103a**. As the first aperture drive ring **120** rotates, the cam pins **104b** rotate the blade members **104** around the rotation pins **104a** along the cam grooves **103c**.

[0035] The cam plate **103** also has cam grooves (cam shape) **103d** with which the cam pins **104b** are engaged, which are evenly provided along the circumferential direction and coaxial with the central axis of the second hole portion **103b**. As the second aperture drive ring **121** rotates, the cam pins **104b** rotate the blade members **104** around the rotation pins **104a** along the cam grooves **103d**.

[0036] Since the plurality of blade members **104** are evenly arranged along the circumferential direction and have the same support structure, the light amount can be adjusted by setting a desired aperture diameter centered on the optical axis of each optical system by rotating the first and second aperture drive rings **120** and **121**. The aperture diameter is adjusted by rotating the blade members **104** by changing a relative phase between the first and second aperture drive rings **120** and **121** and the cam plate **103**.

[0037] In this embodiment, the first and second aperture drive rings **120** and **121** are provided with hole shapes with which the rotation pins **104a** are engaged, and the cam plate **103** is provided with hole shapes (cam grooves) with which the cam pins **104b** are engaged. However, other configurations may be used as long as the blade members **104** rotate by rotating the first and second aperture drive rings **120** and **121** to adjust the light amount. For example, the first and second aperture drive rings **120** and **121** may have hole shapes (cam grooves) with which the cam

pins **104b** are engaged, and the cam plate **103** may have hole shapes with which the rotation pins **104a** are engaged. That is, it is sufficient that one of the first and second aperture drive rings **120** and **121** and the cam plate **103** may have hole shapes with which the rotation pins **104a** are engaged, and the other may have hole shapes (cam grooves) with which the cam pins **104b** are engaged. Also, the aperture base plate **101** may have hole shapes with which the rotation pins **104a** are engaged, and the first and second aperture drive rings **120** and **121** may have hole shapes with which the cam pins **104b** are engaged. Even in this case, it is sufficient that a gap that allows the blade members **104** to rotate may be formed on the opposite side of the first and second aperture drive rings **120** and **121**, with respect to the blade members **104**.

[0038] FIGS. **4A** and **4B** explain the rotating directions of the first and second aperture drive rings **120** and **121** and the phase of the blade members **104**. FIG. **4A** illustrates the light-amount adjusting apparatus **100** in the maximum aperture state, and FIG. **4B** illustrates the positions of the rotation pins **104a** relative to the horizontal direction in this embodiment.

[0039] The optical axis OA, which is the central axis of the first opening **101a**, and the optical axis OB, which is the central axis of the second opening **101b**, are arranged at a 1:1 distance from the central axis of the bearing reference hole **101e** that holds the light-amount adjusting drive unit **105**. The pinion gear **106** is engaged with the first and second ring gears **120b** and **121b** at the same reduction ratio, and the first and second aperture drive rings **120** and **121** can rotate by the same angle in the same rotating direction due to the rotation of the light-amount adjusting drive unit **105**. Thereby, in the first and second optical systems **2** and **3**, the aperture diameters of the optical systems change in synchronization due to the rotation of the light-amount adjusting drive unit **105**, and the light amount can be adjusted.

[0040] There are as many blade members **104** arranged evenly around the optical axis OA, which is the central axis of the first opening **101a**, as there are blade members **104** arranged around the optical axis OB, which is the central axis of the second opening **101b**. The phases of the blade members **104** arranged around the respective optical axes are approximately equal. In this embodiment, six blade members **104** are arranged for each optical system, but the number of blade members **104** is not limited to six.

[0041] As illustrated in FIG. **4B**, the aperture formed by the blade members **104** has a polygonal shape. The phases of the blade members **104** arranged around the respective optical axes are equal. In other words, the phases of the corresponding blade members **104** are the same. Therefore, the angular phases of the polygons formed in the narrowed state are also equal. Thus, the phases (positions in the direction around the optical axis) of the light striations generated when light passes through each optical system are the same. Thereby, in a case where captured images are superimposed and checked using a VR video capture apparatus or the like, the phase shifts of the light striations are the same, and an image that does not provide the unnatural feeling can be obtained. Here, “equal” and “same phase” may be strictly or substantially (approximately) equal or the same phase. A distance between light striations (angle between phases) may have a shift (angle shift, phase shift) of less than 5 degrees from a designed value. That is, the first opening/closing unit and the second opening/closing unit may be arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.

[0042] FIGS. **5A** and **5B** explain the generation principle of light striations generated in a case where light passes through an optical system. FIG. **6** illustrates light striations. FIG. **5A** illustrates the state in which light from a light source passes through the blade members **104**. FIG. **5B** illustrates a light striation generated when the light passes through the blade members **104** and light is irradiated onto the end surface portion of the aperture formed by the blade members **104**. Part of the light that passes through the optical system, as illustrated by a solid line in FIG. **5A**, is shielded

by the blade members **104**, and the other part passes through the aperture. Of the light that passes through the aperture, the light diffracted at the end surface portion of the aperture spreads, wrapping around behind the blade member **104**, as illustrated by a dashed line. The light diffracted at the end surface portion spreads in a band shape outside and inside the end surface portion, as illustrated in FIG. 5B. Thereby, a phenomenon called light striations occurs, as illustrated in FIG. 6. [0043] FIGS. 7A and 7B explain a difference in light striation due to the shape of the aperture. FIG. 7A illustrates a schematic diagram of light striations in a case where the shape of the aperture is an even polygon with an even number of sides. FIG. 7B illustrates a schematic diagram of light striations in a case where the shape of the aperture is an odd polygon with an odd number of sides. [0044] In a case where the shape of the aperture is an even polygon, there are end surface portions facing each other on opposite sides of the optical axis, so the light striations generated at the opposing end faces overlap each other and appear to be a single light striation. In this case, the number of light striations generated is the same as the number of sides of the polygon. In FIG. 7A, the shape of the aperture is hexagonal, so six light rays appear to be radiating from the optical axis. [0045] In a case where the shape of the aperture is an odd polygon, there are no end surfaces facing each other across the optical axis, so twice as many light rays appear to be generated as there are end surface portions. In FIG. 7B, the shape of the aperture is pentagonal, so ten light rays appear to be radiating from the optical axis.

[0046] FIGS. 8A and 8B explain light striations generated in the compound-eyes optical system unit **20** according to the first embodiment. FIG. 8A illustrates light striations generated in each optical system. As illustrated in FIG. 8A, in this embodiment, the light striations generated in each optical system have the same phase. Here, “same” may be strictly or substantially (approximately) the same. FIG. 8B illustrates how the light striations illustrated in FIG. 8A appear in a case where the captured images are superimposed and confirmed using a VR video capture apparatus or the like. As illustrated in FIG. 8B, the light striations generated in the same phase are expressed as almost overlapping.

[0047] As described above, the configuration according to this embodiment can achieve a natural captured image while a good blur effect is obtained.

Second Embodiment

[0048] In the first embodiment, the phases of the blade members **104** arranged for the respective optical systems are the same. However, even if the light striations generated in the respective optical systems are not in the same phase, if the light striations are generated at regular intervals, an image that does not look unnatural can be obtained when the captured images are superimposed and confirmed using a VR video capture apparatus or the like. Regular intervals may be strictly or substantially (approximately) regular intervals.

[0049] This embodiment will discuss a description of how the phases of the blade members **104** arranged for the respective optical systems are shifted to make the intervals between the light striations regular. This embodiment will discuss only the configuration that is different from that of the first embodiment, and will omit a description of the common configuration.

[0050] FIGS. 9A and 9B illustrate example light striations generated in a case where the shape of the aperture is an odd polygon. FIG. 9A illustrates light striations generated in the respective optical systems. In a case where the shape of the aperture is an odd polygon, even if the phase of the aperture is shifted, the phases of the generated light striations are the same as long as the phase shift angle [degrees] of the blade members **104** has a value calculated by the following equation (1) where m is a positive integer:

$$\text{Phase shift angle} = (360 \times m) / (\text{number of blade members } 104 \times 2) \quad (1)$$

[0051] FIG. 9A illustrates the shape of the aperture and the light striations in a case where the shape of the aperture is pentagonal and the phase shift angle is 36 degrees. FIG. 9B illustrates how the light striations in FIG. 9A appear in a case where the captured images are superimposed and

confirmed using a VR video capture apparatus or the like. As illustrated in FIG. 9B, the phases of the light striations generated in the same phase are expressed as almost overlapping.

[0052] FIGS. 10A and 10B illustrate example light striations generated in a case where the shape of the aperture is an even polygon. FIG. 10A illustrates light striations in a case where the shape of the aperture is hexagonal and the phase shift angle is 30 degrees. FIG. 10B illustrates how the light striations in FIG. 10A appear in a case where the captured images are superimposed and confirmed using a VR video capture apparatus or the like. As illustrated in FIG. 10B, the light striations are expressed as overlapping so that they are generated at regular intervals.

[0053] FIGS. 11A and 11B illustrate other example light striations that occur in a case where the shape of the aperture is an odd polygon. FIG. 11A illustrates light striations generated in the respective optical systems in a case where the phase shift angle [degrees] of the blade members 104 has a value calculated by the following equation (2) where m is a positive integer:

$$\text{Phase shift angle} = (360 \times m) / (\text{number of blade members } 104 \times 4) \quad (2)$$

[0054] In FIG. 11A, the shape of the aperture is a pentagon, and the phase shift angle is 18 degrees. FIG. 11B illustrates how the light striations in FIG. 11A appear in a case where the captured images are superimposed and confirmed using a VR video capture apparatus or the like. As illustrated in FIG. 11B, the light striations are expressed as overlapping so that they are generated at regular intervals.

[0055] A distance between adjacent light striations (angle between phases) may have a shift (angle shift, phase shift) of less than 5 degrees from a designed value. That is, the first opening/closing unit and the second opening/closing unit may be arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.

[0056] As described above, the configuration according to this embodiment can achieve a natural captured image while a good blur effect is obtained.

Third Embodiment

[0057] In order to suppress a difference in characteristic such as a difference in aperture diameter accuracy in each optical system, the first and second aperture drive rings 120 and 121 may have the same structure. In a case where a single light-amount adjusting drive unit 105 is provided, it is difficult to align the phases of the plurality of blade members 104 because the assembly phases of the first and second ring gears 120b and 121b relative to the pinion gear 106 are different. This embodiment will discuss a configuration for solving such problems. This embodiment will discuss only the configuration that is different from that of the first embodiment, and will omit the common configuration.

[0058] FIGS. 12A and 12B explain the configuration of the light-amount adjusting apparatus 100 according to this embodiment. FIG. 12A illustrates the light-amount adjusting apparatus 100 in a maximum aperture state, and FIG. 12B illustrates the light-amount adjusting apparatus 100 in a closed state. The first and second aperture drive rings 120 and 121 have the same structure. The first and second ring gears 120b and 121b provided on the first and second aperture drive rings 120 and 121 have gear teeth (cogs) formed in a range wider than an engagement (or geared) range with the pinion gear 106 required from the maximum aperture to a small aperture. Since the first and second aperture drive rings 120 and 121 have the same structure, the gear phases of the first and second ring gears 120b and 121b engaged with the pinion gear 106 and the assembly positional relationship of the respective blade members 104 are the same. Since the first and second aperture drive rings 120 and 121 have the same structure, a rotation amount required for the same light-amount adjusting amount is the same.

[0059] In at least one of the first and second aperture drive rings 120 and 121, at least one of the

first and second ring gears **120b** and **121b** is engaged with the pinion gear **106** so that the phase shift of the blade members **104** reduces. Since the first and second aperture drive rings **120** and **121** have the same structure, the use ranges of the first and second ring gears **120b** and **121b** engaged with the pinion gear **106** during the light amount adjustment are different. This embodiment provides a first drive area that is used for the first aperture drive ring **120** to move from the maximum aperture to a small aperture, and a second drive area that is used for the second aperture drive ring **121** to move from the maximum aperture to a small aperture, which is engaged to match the phase shift of the blade members **104**. Thereby, the phase shifts of the respective blade members **104** can be adjusted to a desired state even if the first and second aperture drive rings **120** and **121** have the same structure.

[0060] As described above, the configuration according to this embodiment can achieve a natural captured image while a good blur effect is obtained.

Fourth Embodiment

[0061] In the configuration described in the third embodiment, the adjustment angle of the engagement between the pinion gear **106** and the first and second ring gears **120b** and **121b** can only be adjusted in units of one gear tooth, so the phases of the respective blade members **104** may not match. This embodiment will discuss a configuration for solving this problem. This embodiment will discuss only the configuration that differs from that of the third embodiment, and will omit a description of the common configuration.

[0062] FIGS. **13A** and **13B** illustrate the configuration of a light-amount adjusting apparatus **100** according to this embodiment. FIG. **13A** illustrates the light-amount adjusting apparatus **100** in the maximum aperture state, and FIG. **13B** illustrates the light-amount adjusting apparatus **100** in a closed state. The first and second aperture drive rings **120** and **121** have the same structure. The first aperture drive ring **120** is provided with a third ring gear (third transmitter) **120d** formed in a different phase in the circumferential direction from that of the first ring gear **120b**. The second aperture drive ring **121** is provided with a fourth ring gear (fourth transmitter) **121d** formed in a different phase in the circumferential direction from that of the second ring gear **121b**. The third and fourth ring gears **120d** and **121d** are formed in an arrangement such that the phases of the blade members **104** for the respective optical systems are the same in a case where the first and fourth ring gears **120b** and **121d** are engaged with the pinion gear **106**. In one of the first and second aperture drive rings **120** and **121**, one of the first and second ring gears **120b** and **121b** is engaged with the pinion gear **106** so as to reduce the phase shift of the blade members **104**. Thereby, the phase shift of the respective blade members **104** can be adjusted to a desired state even if the first and second aperture drive rings **120** and **121** have the same structure.

[0063] As described above, the configuration according to this embodiment can achieve a natural captured image while a good blur effect is obtained.

[0064] While the disclosure has described example embodiments, it is to be understood that the disclosure is not limited to the example embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0065] The configuration according to each embodiment can achieve a natural captured image while a good blur effect is obtained.

[0066] This application claims priority to Japanese Patent Application No. 2024-035996, which was filed on Mar. 8, 2024, and which is hereby incorporated by reference herein in its entirety.

Claims

1. A light-amount adjusting apparatus configured to adjust a light amount of a first optical system and a light amount of a second optical system, the light-amount adjusting apparatus comprising: a drive unit configured to drive blade members; a first opening/closing unit including three or more

blade members and configured to open and close at least a part of a first opening through which light is to pass, according to driving of the drive unit; and a second opening/closing unit including blade members as many as the blade members constituting the first opening/closing unit and configured to open and close at least a part of a second opening through which light is to pass, according to driving of the drive unit, wherein the first opening/closing unit and the second opening/closing unit are arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.

2. The light-amount adjusting apparatus according to claim 1, wherein the first opening/closing unit and the second opening/closing unit are arranged so that corresponding blade members are in the same phase.

3. The light-amount adjusting apparatus according to claim 1, wherein in a case where the number of the blade members constituting each of the first opening/closing unit and the second opening/closing unit is an odd number, the first opening/closing unit and the second opening/closing unit are arranged so that phases of positions of corresponding blade members are shifted by $(360 \times m) / (\text{the number of the blade members constituting each of the first opening/closing unit and the second opening/closing unit} \times 4)$ [degrees] (where m is a positive integer).

4. The light-amount adjusting apparatus according to claim 1, wherein in a case where the number of the blade members constituting each of the first opening/closing unit and the second opening/closing unit is an even number, the first opening/closing unit and the second opening/closing unit are arranged so that the phases of the positions of the corresponding blade members are shifted by $(360 \times m) / (\text{the number of the blade members constituting each of the first opening/closing unit and the second opening/closing unit} \times 2)$ [degrees] (where m is a positive integer).

5. The light-amount adjusting apparatus according to claim 1, further comprising: a transmission member configured to transmit a drive force from a drive unit, a first drive member rotatably engaged with the first opening; and a second drive member rotatably engaged with the second opening, wherein the first drive member includes a first transmitter configured to receive the drive force via the transmission member and rotate according to the driving of the drive unit, wherein the second drive member includes a second transmitter configured to receive the drive force via the transmission member and rotate according to the driving of the drive unit, wherein the blade members constituting the first opening/closing unit are evenly arranged radially around a center of a rotation axis of the first drive member, and wherein the blade members constituting the second opening/closing unit are evenly arranged radially around a center of a rotation axis of the second drive member.

6. The light-amount adjusting apparatus according to claim 5, wherein a gap that allows the blade members to rotate is formed on an opposite side of the first drive member and the second drive member with respect to the blade members.

7. The light-amount adjusting apparatus according to claim 5, wherein the first drive member and the second drive member have the same structure, and wherein an area of the first transmitter that is used for rotation of the first drive member is different from an area of the second transmitter that is used for rotation of the second drive member.

8. The light-amount adjusting apparatus according to claim 5, wherein the first drive member further includes a third transmitter provided in a different phase from the first transmitter and configured to receive the drive force via the transmission member, wherein the second drive member further includes a fourth transmitter provided in a different phase from the second transmitter and configured to receive the drive force via the transmission member, wherein the first drive member and the second drive members have the same structure, and wherein the first drive

member receives the drive force via the first transmitter and the second drive member receives the drive force via the fourth transmitter.

9. The light-amount adjusting apparatus according to claim 5, further comprising: a base member holding the drive unit and having the first opening and the second opening; and a press member fixed to the base member, wherein the first opening is disposed with a center on an optical axis of the first optical system, wherein the second opening is disposed with a center on an optical axis of the second optical system, wherein each of the blade members includes a first engagement portion that serves as a rotation center and a second engagement portion for rotating the blade members, wherein the first engagement portion is engaged with a hole portion formed in one of a pair of the first drive member and the second drive member, and the press member, and wherein the second engagement portion is engaged with a cam shape formed in another of the pair of the first drive member and the second drive member, and the press member.

10. The light-amount adjusting apparatus according to claim 9, wherein the light amount of the first optical system is adjusted by adjusting a relative phase between the press member and the first drive member, and by rotating the blade members constituting the first opening/closing unit, and wherein the light amount of the second optical system is adjusted by adjusting a relative phase between the press member and the second driving member, and by rotating the blade members constituting the second opening/closing unit.

11. The light-amount adjusting apparatus according to claim 1, further comprising: a base member holding the drive unit and having the first opening and the second opening, wherein the first opening is disposed with a center on an optical axis of the first optical system, and wherein the second opening is disposed with a center on an optical axis of the second optical system.

12. A lens apparatus comprising: a first optical system; a second optical system; and a light-amount adjusting apparatus configured to adjust a light amount of the first optical system and a light amount of the second optical system, wherein the light amount adjusting apparatus includes: a drive unit configured to drive blade members; a first opening/closing unit including three or more blade members and configured to open and close at least a part of a first opening through which light is to pass, according to driving of the drive unit; and a second opening/closing unit including blade members as many as the blade members constituting the first opening/closing unit and configured to open and close at least a part of a second opening through which light is to pass, according to driving of the drive unit, wherein the first opening/closing unit and the second opening/closing unit are arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.

13. An image pickup apparatus comprising: a light-amount adjusting apparatus configured to adjust a light amount of a first optical system and a light amount of a second optical system; and an image sensor configured to photoelectrically convert an optical image formed by at least one of the first optical system and the second optical system, wherein the light-amount adjusting apparatus includes: a drive unit configured to drive blade members; a first opening/closing unit including three or more blade members and configured to open and close at least a part of a first opening through which light is to pass, according to driving of the drive unit; and a second opening/closing unit including blade members as many as the blade members constituting the first opening/closing unit and configured to open and close at least a part of a second opening through which light is to pass, according to driving of the drive unit, wherein the first opening/closing unit and the second opening/closing unit are arranged so that in a case where images obtained by imaging using the first optical system and the second optical system are superimposed and all angles between adjacent light striations generated when the light passes through apertures formed by the first

opening/closing unit and the second opening/closing unit are equal angles, all the angles between adjacent light striations fall within ± 5 degrees.
